ORGANIC TURMERIC PRODUCTION FOR GERMAN MARKET STANDARD MANUAL

ORGTURM INVESTMENTS Private Limited

A subsidiary of Global African Herbal Solutions (GAHS)

BACKGROUND

Curcuma longa is commonly known as turmeric. It has several varieties of species which we are mainly producing Lakadong Turmeric which has a high curcumin content of 6-7.5 %. The crop is a rhizomatous herbaceous perennial plant in the family (Zingiberaceae) of the ginger plant. It is a flowering plant native to India producing white and yellow to purple colours. Turmeric primarily through its active compound curcumin, offers potent anti-inflammatory, antioxidant, and antimicrobial benefits. It supports joint health, eases arthritis pain, improves digestion, and acts as a cardio protective agent. It also promotes brain health, aids in weight loss, and improves skin conditions, anti-aging and a lot more medicinal benefits.

The post covid era has therefore led to a very sharp rise on demand for what we are now introducing as organically produced turmeric strictly for our German pharmaceutical industry where the following document seeks to outline the proper organic turmeric crop production management.

SITE SELECTION AND CLIMATE

It is imperative to note that first thing first soil samples collection and testing for the certification of the soil is done for any selected site such that it is proven if it is free from any traces of agro-synthetic chemical compounds by scientifically tested data which is to be done only by Africa University.

Soil samples collection point GPS coordinates are picked up by assigned specialists for Ecocert Certification and sent to Germany for the record keeping and booking of the farm in the market data base this is also practically used for monitoring and certification of the farm activities and operations.

Buffer zones of 200 to 300m are identified and marked where there is a potential risk of agro synthetic chemical compounds contamination from adjacent activities.

Detailing on slope, insect pest with pesticide, herbicides, insecticides as well as wind drift management.

Certification continues throughout the 7 to 9 months growing season where our team continually and randomly make visits to all our valued committed farmers. Documentation of operational activities is done at farm level to which traceability checks are made reference to.

The best soil types for the production of organic turmeric would require well drained sandy loam soils of pH 4.5 - 7.5 and thriving in temperatures averaging 20 - 35 deg C. These soils should be naturally virgin or land that has been laid farrow for up to 2-3 years free from any agro- synthetic chemicals contamination ranging from common AN fertilizer, pesticides, insecticides and herbicides.

LAND PREPARATION AND PLANTING

The Relict Genesis Liquid Fertilizer is applied to the soil at a rate of 2 to 3 l per ha to a total of 1000l and sprayed to the soil before planting. It acts as a liming substance and basal fertilizer which fixes and prepares the fertility trenches for the crop to effectively access and uptake the nutrients from the seed bed.

The standard acceptable beds would be 1.6m wide which effective planting area would be *1.4m* and 30cm deep spaced at 40cm cat walk working space which are either high raised or trenches.

The beds or fertility trenches are fed with organic animal manure , cow dung, goat manure, chicken manure at a rate of 20 to 30t per ha.

These trenches are laid for up to 2 months to allow the manure to bio degrade so as to be able to sustain plant growth by supplying the necessary nutrients to the crop.

Vegetative mulch, green or dead to cover the first leaves of the tubers after planting to provide for a sound shade avoiding too much direct heat sun rays on the tender growing auxiliary leaflet.

Nutrient organic basal setting which would be consistently backed up by our Relict Genesis Liquid Fertilizer.

Seed is also treated with Relict p before planting at a rate of 500mls per 200l soaked overnight or make sprays enough to wet all the seed before planting and kept in a dark room for up to 2 weeks for effective sprouting.

The seedlings after sprouting are carefully cut into individual seedlings which are effectively planted at an effective spacing of 20cm x 20cm on the fertility trenches assumed at 100m x 1.4mx 0.3m deep beds making a total of 50 beds.

For plant count we only use width x length

Plants along the length

100m/0.2m = 501 planting stations

Total plants across the width bed

1.4m / 0.2m = 8 planting stations

8 x 501 = 4008 plants per bed.

Drip irrigation system are effectively recommended for this project.

Spacing : 0.20m x 0.20m = 0.04 m2/plant

Bed area : 100m x 1.4m = 140 m2/bed (ignoring cat walk for now)

Total bed area : 50 beds x 140 m2 = 8000 m2 = 0.8 ha

Plants per bed : 140 m2/ = 4008 plants

Plant population : 8000 m2 / = 200 400 plants per ha where cat walk of 0.4m has been factored out already .

It is also practically assumed that effectively 200 400 plants will definitely be produced all things managed to standard where a provision of 10% for mortality rate can be factored in giving us a total of 180 360 plants after 7 - 9 months of production.

Proper management has proven to give a productivity rate per ha of up to more than 0.5kg to 1.5kg per plant which gives us an estimated total yield of 180 000kg as the maximum possible and average minimum of nothing less than 40 000kg selling at a minimum of nothing less than 5EUR to 20EUR per kg.

A marking for the pitting box has been designed 1m x 1.4m for the smooth management of the seed population per ha which gives us 48 pits per each box where a farmer will not adopt the drip irrigation system installation services.

IRRIGATION

Borehole water has been used as recommended irrigation water where testing for the water is also recommended especially for those opting open water sources. Drip irrigation system installation is also practically recommended as well.

Filters for too salty borehole water are used to mitigate the salt water from entering the field which compromises the optimum growth of the crop, and could also have a clogging effect on the drip tape holes.

WEEDING AND FERTILIZER APPLICATION

Weeding is to be done manually twice a month after every 2 weeks and so shall be Relict Genesis P Organic Liquid Fertilizer to be applied at a rate of 50mls per 20l which is 500ml per ha per 2wks and 1l per month.

HARVESTING AND PACKAGING

Harvesting is done using garden hand forks and produce is carefully cleaned with water and air dried under our grading and packaging storage facility. The produce is packed up into 25 to 50kg hessian bags which are then weighed and scaled using an electric scale for establishing total actual yield.

Shipment plan is therefore put into action where half of the total yield of the farmer is paid upfront before movement of the consignment to Germany after samples have qualified.

For the National Project Operations Coordinator

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